10. Interpolate to find Y_2.
    - The two closest F_a/C_0 values in the table will be the x-points. The two associated Y_2 values will be the y-points.
    - Calculate the slope: $m = \frac{y_1 - y_2}{x_1 - x_2}$
    - Plug in a point into $y = mx + b$ and solve for b
    - Finally, plug in your value of F_a/C_0 as x to get Y_2 as y
11. Recalculate F_e with the new Y_2 value:

$$F_e = X_i V F_r + Y_i F_a$$

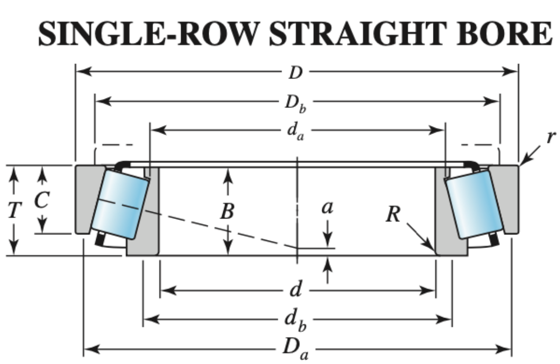
12. Calculate the new C_{10}. The calculation for C_{10} only changes in F_e so we can do the following calculation:

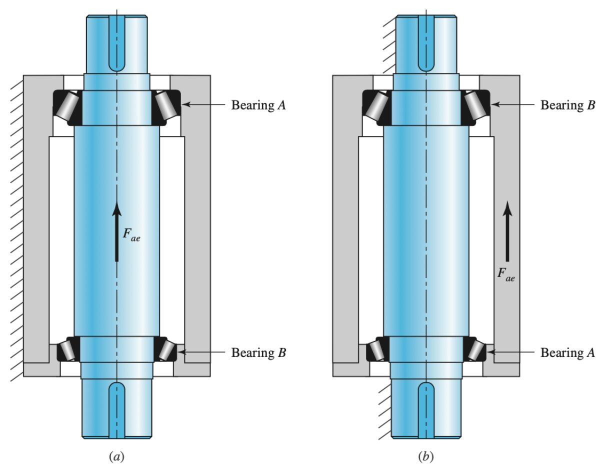
$$\text{new } C_{10} = \frac{\text{new } F_e}{\text{old } F_e}(\text{old } C_{10})$$

13. Select a bearing that has a C_{10} value that exceeds the calculated one. Refer to tables 11-2 or 11-3 from **step 8** of the previous section.
14. Repeat **steps 6 to 13** until you select the same bearing from the table twice in a row.

4.3 Tapered Roller Bearings

4.3.1 Anatomy





4.3.2 Design Selection

1. Calculate the radial and axial loads on each bearing
    - $F_{rA} = \sqrt{R_{xA}^2 + R_{yA}^2}$
    - $F_{rB} = \sqrt{R_{xB}^2 + R_{yB}^2}$

- Assume bearing A carries the axial load: $F_{ae} = F_a$

2. Calculate the induced loads for each bearings. For our initial calculation we will assume $K_A = K_B = 1.5$.

$$F_{iA} = \frac{0.47F_{rA}}{K_A}$$
$$F_{iB} = \frac{0.47F_{rB}}{K_B}$$

3. Calculate the equivalent loads.

$$\text{If} \quad F_{iA} \le (F_{iB} + F_{ae}) \quad \begin{cases} F_{eA} = 0.4F_{rA} + K_A(F_{iB} + F_{ae}) \\ F_{eB} = F_{rB} \end{cases}$$

$$\text{If} \quad F_{iA} > (F_{iB} + F_{ae}) \quad \begin{cases} F_{eB} = 0.4F_{rB} + K_B(F_{iA} - F_{ae}) \\ F_{eA} = F_{rA} \end{cases}$$

4. Get bearing design life $\mathscr{L}_D$ for your application from this table (if not given in the question):

Table 11–4 Bearing-Life Recommendations for Various Classes of Machinery

Type of Application	Life, kh
Instruments and apparatus for infrequent use	Up to 0.5
Aircraft engines	0.5–2
Machines for short or intermittent operation where service interruption is of minor importance	4–8
Machines for intermittent service where reliable operation is of great importance	8–14
Machines for 8-h service that are not always fully utilized	14–20
Machines for 8-h service that are fully utilized	20–30
Machines for continuous 24-h service	50–60
Machines for continuous 24-h service where reliability is of extreme importance	100–200

5. Calculate the multiple of rating life for each bearing in the assembly:

$$x_D = \frac{L_D}{L_{10}} = \frac{60\mathscr{L}_D n_D}{L_{10}}$$

where L_D is bearing design life in number of revolutions,
L_{10} is the rating life,
$\mathscr{L}_D$ is the design life in hours,
n_D is the angular speed of the bearing in rpm

The Weibull parameters used will depend on which manufacturer's bearings we are using.

- Timken (Manufacturer 1) is common for tapered roller bearings
- SKF (Manufacturer 2) is common for ball and straight roller bearings

		Weibull Parameters Rating Lives		
Manufacturer	Rating Life, Revolutions	X_0	θ	b
1	$90(10^6)$	0	4.48	1.5
2	$1(10^6)$	0.02	4.459	1.483

6. Calculate the bearing reliability of each individual bearing (if the total reliability of the ensemble is given):

$$R_i = \sqrt[n]{R_{tot}}$$

where R_i is the individual bearing reliability,
R_{tot} is the total bearing reliability,
n is the number of bearings in the assembly

7. For each bearing, calculate the load rating C_{10} for the equivalent load:

$$C_{10} = a_f F_D \left[\frac{x_D}{x_0 + (\theta - x_0)[\ln(1/R_D)]^{1/b}} \right]^{1/a}$$

- $a = 10/3$ for tapered roller bearings

8. Select tentative tapered roller bearings from the following table that have one-row radial values that exceed the calculated C_{10} values:

									cone				cup			
bore	outside diameter	width	rating at 500 rpm for 3000 hours L_{10} one-row radial	thrust	factor	eff. load center	part numbers cone	cup	max shaft fillet radius	width	backing shoulder diameters		max housing fillet radius	width	backing shoulder diameters	
d	**D**	**T**	**N** **lbf**	**N** **lbf**	**K**	**a**②			**R**①	**B**	d_b	d_a	**r**①	**C**	D_b	D_a
25.000 0.9843	**52.000** 2.0472	**16.250** 0.6398	**8190** 1840	**5260** 1180	**1.56**	**–3.6** –0.14	**◆30205**	**◆30205**	**1.0** 0.04	**15.000** 0.5906	**30.5** 1.20	**29.0** 1.14	**1.0** 0.04	**13.000** 0.5118	**46.0** 1.81	**48.5** 1.91
25.000 0.9843	**52.000** 2.0472	**19.250** 0.7579	**9520** 2140	**9510** 2140	**1.00**	**–3.0** –0.12	**◆32205-B**	**◆32205-B**	**1.0** 0.04	**18.000** 0.7087	**34.0** 1.34	**31.0** 1.22	**1.0** 0.04	**15.000** 0.5906	**43.5** 1.71	**49.5** 1.95
25.000 0.9843	**52.000** 2.0472	**22.000** 0.8661	**13200** 2980	**7960** 1790	**1.66**	**–7.6** –0.30	**◆33205**	**◆33205**	**1.0** 0.04	**22.000** 0.8661	**34.0** 1.34	**30.5** 1.20	**1.0** 0.04	**18.000** 0.7087	**44.5** 1.75	**49.0** 1.93
25.000 0.9843	**62.000** 2.4409	**18.250** 0.7185	**13000** 2930	**6680** 1500	**1.95**	**–5.1** –0.20	**◆30305**	**◆30305**	**1.5** 0.06	**17.000** 0.6693	**32.5** 1.28	**30.0** 1.18	**1.5** 0.06	**15.000** 0.5906	**55.0** 2.17	**57.0** 2.24
25.000 0.9843	**62.000** 2.4409	**25.250** 0.9941	**17400** 3910	**8930** 2010	**1.95**	**–9.7** –0.38	**◆32305**	**◆32305**	**1.5** 0.06	**24.000** 0.9449	**35.0** 1.38	**31.5** 1.24	**1.5** 0.06	**20.000** 0.7874	**54.0** 2.13	**57.0** 2.24
25.159 0.9905	**50.005** 1.9687	**13.495** 0.5313	**6990** 1570	**4810** 1080	**1.45**	**–2.8** –0.11	**07096**	**07196**	**1.5** 0.06	**14.260** 0.5614	**31.5** 1.24	**29.5** 1.16	**1.0** 0.04	**9.525** 0.3750	**44.5** 1.75	**47.0** 1.85
25.400 1.0000	**50.005** 1.9687	**13.495** 0.5313	**6990** 1570	**4810** 1080	**1.45**	**–2.8** –0.11	**07100**	**07196**	**1.0** 0.04	**14.260** 0.5614	**30.5** 1.20	**29.5** 1.16	**1.0** 0.04	**9.525** 0.3750	**44.5** 1.75	**47.0** 1.85
25.400 1.0000	**50.005** 1.9687	**13.495** 0.5313	**6990** 1570	**4810** 1080	**1.45**	**–2.8** –0.11	**07100-S**	**07196**	**1.5** 0.06	**14.260** 0.5614	**31.5** 1.24	**29.5** 1.16	**1.0** 0.04	**9.525** 0.3750	**44.5** 1.75	**47.0** 1.85
25.400 1.0000	**50.292** 1.9800	**14.224** 0.5600	**7210** 1620	**4620** 1040	**1.56**	**–3.3** –0.13	**L44642**	**L44610**	**3.5** 0.14	**14.732** 0.5800	**36.0** 1.42	**29.5** 1.16	**1.3** 0.05	**10.668** 0.4200	**44.5** 1.75	**47.0** 1.85
25.400 1.0000	**50.292** 1.9800	**14.224** 0.5600	**7210** 1620	**4620** 1040	**1.56**	**–3.3** –0.13	**L44643**	**L44610**	**1.3** 0.05	**14.732** 0.5800	**31.5** 1.24	**29.5** 1.16	**1.3** 0.05	**10.668** 0.4200	**44.5** 1.75	**47.0** 1.85
25.400 1.0000	**51.994** 2.0470	**15.011** 0.5910	**6990** 1570	**4810** 1080	**1.45**	**–2.8** –0.11	**07100**	**07204**	**1.0** 0.04	**14.260** 0.5614	**30.5** 1.20	**29.5** 1.16	**1.3** 0.05	**12.700** 0.5000	**45.0** 1.77	**48.0** 1.89
25.400 1.0000	**56.896** 2.2400	**19.368** 0.7625	**10900** 2450	**5740** 1290	**1.90**	**–6.9** –0.27	**1780**	**1729**	**0.8** 0.03	**19.837** 0.7810	**30.5** 1.20	**30.0** 1.18	**1.3** 0.05	**15.875** 0.6250	**49.0** 1.93	**51.0** 2.01
25.400 1.0000	**57.150** 2.2500	**19.431** 0.7650	**11700** 2620	**10900** 2450	**1.07**	**–3.0** –0.12	**M84548**	**M84510**	**1.5** 0.06	**19.431** 0.7650	**36.0** 1.42	**33.0** 1.30	**1.5** 0.06	**14.732** 0.5800	**48.5** 1.91	**54.0** 2.13
25.400 1.0000	**58.738** 2.3125	**19.050** 0.7500	**11600** 2610	**6560** 1470	**1.77**	**–5.8** –0.23	**1986**	**1932**	**1.3** 0.05	**19.355** 0.7620	**32.5** 1.28	**30.5** 1.20	**1.3** 0.05	**15.080** 0.5937	**52.0** 2.05	**54.0** 2.13
25.400 1.0000	**59.530** 2.3437	**23.368** 0.9200	**13900** 3140	**13000** 2930	**1.07**	**–5.1** –0.20	**M84249**	**M84210**	**0.8** 0.03	**23.114** 0.9100	**36.0** 1.42	**32.5** 1.27	**1.5** 0.06	**18.288** 0.7200	**49.5** 1.95	**56.0** 2.20
25.400 1.0000	**60.325** 2.3750	**19.842** 0.7812	**11000** 2480	**6550** 1470	**1.69**	**–5.1** –0.20	**15578**	**15523**	**1.3** 0.05	**17.462** 0.6875	**32.5** 1.28	**30.5** 1.20	**1.5** 0.06	**15.875** 0.6250	**51.0** 2.01	**54.0** 2.13
25.400 1.0000	**61.912** 2.4375	**19.050** 0.7500	**12100** 2730	**7280** 1640	**1.67**	**–5.8** –0.23	**15101**	**15243**	**0.8** 0.03	**20.638** 0.8125	**32.5** 1.28	**31.5** 1.24	**2.0** 0.08	**14.288** 0.5625	**54.0** 2.13	**58.0** 2.28
25.400 1.0000	**62.000** 2.4409	**19.050** 0.7500	**12100** 2730	**7280** 1640	**1.67**	**–5.8** –0.23	**15100**	**15245**	**3.5** 0.14	**20.638** 0.8125	**38.0** 1.50	**31.5** 1.24	**1.3** 0.05	**14.288** 0.5625	**55.0** 2.17	**58.0** 2.28
25.400 1.0000	**62.000** 2.4409	**19.050** 0.7500	**12100** 2730	**7280** 1640	**1.67**	**–5.8** –0.23	**15101**	**15245**	**0.8** 0.03	**20.638** 0.8125	**32.5** 1.28	**31.5** 1.24	**1.3** 0.05	**14.288** 0.5625	**55.0** 2.17	**58.0** 2.28

									cone				cup			
bore	outside diameter	width	rating at 500 rpm for 3000 hours L_{10}		fac-tor	eff. load center	part numbers		max shaft fillet radius	width	backing shoulder diameters		max hous-ing fillet radius	width	backing shoulder diameters	
			one-row radial	thrust			cone	cup								
d	D	T	N lbf	N lbf	K	a②			R①	B	d_b	d_a	r①	C	D_b	D_a
25.400 1.0000	62.000 2.4409	19.050 0.7500	12100 2730	7280 1640	1.67	−5.8 −0.23	15102	15245	1.5 0.06	20.638 0.8125	34.0 1.34	31.5 1.24	1.3 0.05	14.288 0.5625	55.0 2.17	58.0 2.28
25.400 1.0000	62.000 2.4409	20.638 0.8125	12100 2730	7280 1640	1.67	−5.8 −0.23	15101	15244	0.8 0.03	20.638 0.8125	32.5 1.28	31.5 1.24	1.3 0.05	15.875 0.6250	55.0 2.17	58.0 2.28
25.400 1.0000	63.500 2.5000	20.638 0.8125	12100 2730	7280 1640	1.67	−5.8 −0.23	15101	15250	0.8 0.03	20.638 0.8125	32.5 1.28	31.5 1.24	1.3 0.05	15.875 0.6250	56.0 2.20	59.0 2.32
25.400 1.0000	63.500 2.5000	20.638 0.8125	12100 2730	7280 1640	1.67	−5.8 −0.23	15101	15250X	0.8 0.03	20.638 0.8125	32.5 1.28	31.5 1.24	1.5 0.06	15.875 0.6250	55.0 2.17	59.0 2.32
25.400 1.0000	64.292 2.5312	21.433 0.8438	14500 3250	13500 3040	1.07	−3.3 −0.13	M86643	M86610	1.5 0.06	21.433 0.8438	38.0 1.50	36.5 1.44	1.5 0.06	16.670 0.6563	54.0 2.13	61.0 2.40
25.400 1.0000	65.088 2.5625	22.225 0.8750	13100 2950	16400 3690	0.80	−2.3 −0.09	23100	23256	1.5 0.06	21.463 0.8450	39.0 1.54	34.5 1.36	1.5 0.06	15.875 0.6250	53.0 2.09	63.0 2.48
25.400 1.0000	66.421 2.6150	23.812 0.9375	18400 4140	8000 1800	2.30	−9.4 −0.37	2687	2631	1.3 0.05	25.433 1.0013	33.5 1.32	31.5 1.24	1.3 0.05	19.050 0.7500	58.0 2.28	60.0 2.36
25.400 1.0000	68.262 2.6875	22.225 0.8750	15300 3440	10900 2450	1.40	−5.1 −0.20	02473	02420	0.8 0.03	22.225 0.8750	34.5 1.36	33.5 1.32	1.5 0.06	17.462 0.6875	59.0 2.32	63.0 2.48
25.400 1.0000	72.233 2.8438	25.400 1.0000	18400 4140	17200 3870	1.07	−4.6 −0.18	HM88630	HM88610	0.8 0.03	25.400 1.0000	39.5 1.56	39.5 1.56	2.3 0.09	19.842 0.7812	60.0 2.36	69.0 2.72
25.400 1.0000	72.626 2.8593	30.162 1.1875	22700 5110	13000 2910	1.76	−10.2 −0.40	3189	3120	0.8 0.03	29.997 1.1810	35.5 1.40	35.0 1.38	3.3 0.13	23.812 0.9375	61.0 2.40	67.0 2.64
26.157 1.0298	62.000 2.4409	19.050 0.7500	12100 2730	7280 1640	1.67	−5.8 −0.23	15103	15245	0.8 0.03	20.638 0.8125	33.0 1.30	32.5 1.28	1.3 0.05	14.288 0.5625	55.0 2.17	58.0 2.28
26.162 1.0300	63.100 2.4843	23.812 0.9375	18400 4140	8000 1800	2.30	−9.4 −0.37	2682	2630	1.5 0.06	25.433 1.0013	34.5 1.36	32.0 1.26	0.8 0.03	19.050 0.7500	57.0 2.24	59.0 2.32
26.162 1.0300	66.421 2.6150	23.812 0.9375	18400 4140	8000 1800	2.30	−9.4 −0.37	2682	2631	1.5 0.06	25.433 1.0013	34.5 1.36	32.0 1.26	1.3 0.05	19.050 0.7500	58.0 2.28	60.0 2.36
26.975 1.0620	58.738 2.3125	19.050 0.7500	11600 2610	6560 1470	1.77	−5.8 −0.23	1987	1932	0.8 0.03	19.355 0.7620	32.5 1.28	31.5 1.24	1.3 0.05	15.080 0.5937	52.0 2.05	54.0 2.13
† 26.988 † 1.0625	50.292 1.9800	14.224 0.5600	7210 1620	4620 1040	1.56	−3.3 −0.13	L44649	L44610	3.5 0.14	14.732 0.5800	37.5 1.48	31.0 1.22	1.3 0.05	10.668 0.4200	44.5 1.75	47.0 1.85
† 26.988 † 1.0625	60.325 2.3750	19.842 0.7812	11000 2480	6550 1470	1.69	−5.1 −0.20	15580	15523	3.5 0.14	17.462 0.6875	38.5 1.52	32.0 1.26	1.5 0.06	15.875 0.6250	51.0 2.01	54.0 2.13
† 26.988 † 1.0625	62.000 2.4409	19.050 0.7500	12100 2730	7280 1640	1.67	−5.8 −0.23	15106	15245	0.8 0.03	20.638 0.8125	33.5 1.32	33.0 1.30	1.3 0.05	14.288 0.5625	55.0 2.17	58.0 2.28
† 26.988 † 1.0625	66.421 2.6150	23.812 0.9375	18400 4140	8000 1800	2.30	−9.4 −0.37	2688	2631	1.5 0.06	25.433 1.0013	35.0 1.38	33.0 1.30	1.3 0.05	19.050 0.7500	58.0 2.28	60.0 2.36
28.575 1.1250	56.896 2.2400	19.845 0.7813	11600 2610	6560 1470	1.77	−5.8 −0.23	1985	1930	0.8 0.03	19.355 0.7620	34.0 1.34	33.5 1.32	0.8 0.03	15.875 0.6250	51.0 2.01	54.0 2.11
28.575 1.1250	57.150 2.2500	17.462 0.6875	11000 2480	6550 1470	1.69	−5.1 −0.20	15590	15520	3.5 0.14	17.462 0.6875	39.5 1.56	33.5 1.32	1.5 0.06	13.495 0.5313	51.0 2.01	53.0 2.09
28.575 1.1250	58.738 2.3125	19.050 0.7500	11600 2610	6560 1470	1.77	−5.8 −0.23	1985	1932	0.8 0.03	19.355 0.7620	34.0 1.34	33.5 1.32	1.3 0.05	15.080 0.5937	52.0 2.05	54.0 2.13
28.575 1.1250	58.738 2.3125	19.050 0.7500	11600 2610	6560 1470	1.77	−5.8 −0.23	1988	1932	3.5 0.14	19.355 0.7620	39.5 1.56	33.5 1.32	1.3 0.05	15.080 0.5937	52.0 2.05	54.0 2.13
28.575 1.1250	60.325 2.3750	19.842 0.7812	11000 2480	6550 1470	1.69	−5.1 −0.20	15590	15523	3.5 0.14	17.462 0.6875	39.5 1.56	33.5 1.32	1.5 0.06	15.875 0.6250	51.0 2.01	54.0 2.13
28.575 1.1250	60.325 2.3750	19.845 0.7813	11600 2610	6560 1470	1.77	−5.8 −0.23	1985	1931	0.5 0.03	19.355 0.7620	34.0 1.34	33.5 1.32	1.3 0.05	15.875 0.6250	52.0 2.05	55.0 2.17

9. Obtain new K_A and K_B values and repeat **steps 2 to 8** until you select the same bearing from the table twice in a row.

Note that the calculation for C_{10} only changes in F_e so we can use the following to calculate the new C_{10} value in every iteration:

$$\text{new } C_{10} = \frac{\text{new } F_e}{\text{old } F_e}(\text{old } C_{10})$$

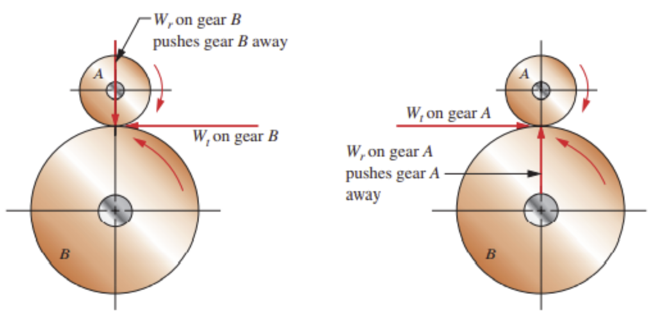
4.4 Shafts and Keys

4.4.1 Design Selection

1. You should be given some shaft with length dimensions and certain gears/sheaves attached as well as bearing locations.
   Find the torque from each sheave/gear from the power (P) and angular speed (n)
   $$T = \frac{63000P}{n}$$
   Verify that the sum of torques is equal to 0

2. Compute the force from each sheave/gear

   (a) For spur gears:
   $$\text{tangential force: } W_t = \frac{2T}{D}$$
   $$\text{radial force: } W_r = W_t \tan\phi$$
   $D =$ pitch diameter
   $\phi =$ pressure angle
   The direction will be given by



(*a*) ***Action forces***—gear *A* drives gear *B*. Forces exerted on gear *B* by gear *A*.

(*b*) ***Reaction forces.*** Forces exerted on gear *A* by gear *B*.

   (b) For helical gears:
   $$\text{tangential force: } W_t = \frac{2T}{D}$$
   $$\text{radial force: } W_r = W_t\frac{\tan\phi_n}{\cos\psi}$$
   $$\text{axial force: } W_x = W_t \tan\psi$$
   $\phi_n =$ normal pressure angle
   $\psi =$ helix angle